NBA Shot Make Probability Project

Project Objective

The objective of this project was to develop a model that estimates the probability that an NBA shot will be made using information available about the shot, shooter, defenders, location, and game situation.

Rather than predicting only a binary make/miss outcome, the model produces a probability between 0 and 1 for every test shot. This allows the predictions to represent differences in shot difficulty and provides a probabilistic estimate that can be evaluated using log-loss.

The final model was trained on 425,719 historical shots and generated predictions for 213,977 test shots.

Data Exploration

The training dataset contains shot-level information covering several categories of basketball context.

Shot and Location Information

Important variables include:

- Shot type
- Shot distance
- Shot location (`locationx`, `locationy`)
- Whether the attempt was a three-point shot
- Number of dribbles before the shot
- Time remaining on the shot clock

Defensive Information

The dataset also contains information about defensive pressure, including:

- Closest defender distance
- Number of contesting defenders
- Distances to individual contesters
- Whether the shot was contested
- Closest defender approach information at multiple points before release

The provided defensive tracking data records the closest defender's distance at 1.0, 0.75, 0.5, and 0.25 seconds before the shot.

Game and Player Information

Additional variables describe:

- Shooter
- Offensive team
- Defensive team
- Game state
- Month
- Closest defender

The training data contained **195,009 made shots** and **230,710 missed shots**.

I also examined the distribution of categorical and numerical variables and verified that the numerical variables used in modeling did not contain missing values.

Feature Engineering

I created additional features to give the model more useful information about shot difficulty and defensive pressure.

One focus was defensive tracking. The closest defender's approach data was transformed into individual features representing defensive distance at different points before the shot, including approximately 1 second, 0.5 seconds, and 0.25 seconds before release.

These features allow the model to account not only for how close a defender was at release, but also for how the defensive pressure developed immediately before the shot.

I also created nonlinear distance features, including squared shot distance, to allow the model to capture relationships between distance and shooting probability that may not be linear.

Additional engineered features were used to represent relationships within the existing shot and location information.

Baseline Model

I first established a baseline model to provide a reference point for evaluating subsequent improvements.

The baseline model achieved a validation log-loss of:

0.65618

Log-loss was selected as the primary evaluation metric because the goal is to produce accurate probabilities rather than simply classify each shot as a make or miss.

A model that predicts a 0.80 probability for a shot is making a substantially different prediction from one that predicts 0.55, so the evaluation metric needs to account for the quality and confidence of the probabilities.

Lower log-loss represents better probabilistic predictions.

Model Development

I progressively incorporated additional features and model complexity to determine whether they improved predictive performance.

An improved model using additional shot and contextual information reduced validation log-loss to:

0.64317

I then incorporated defensive-tracking information, which further improved validation performance to approximately:

0.63997

This improvement suggested that information about defensive proximity and movement contained additional predictive information beyond basic shot characteristics.

For the final model, I used CatBoost, a gradient-boosting algorithm that can effectively model nonlinear relationships and interactions among numerical and categorical variables.

The final CatBoost model achieved a validation log-loss of:

0.62084

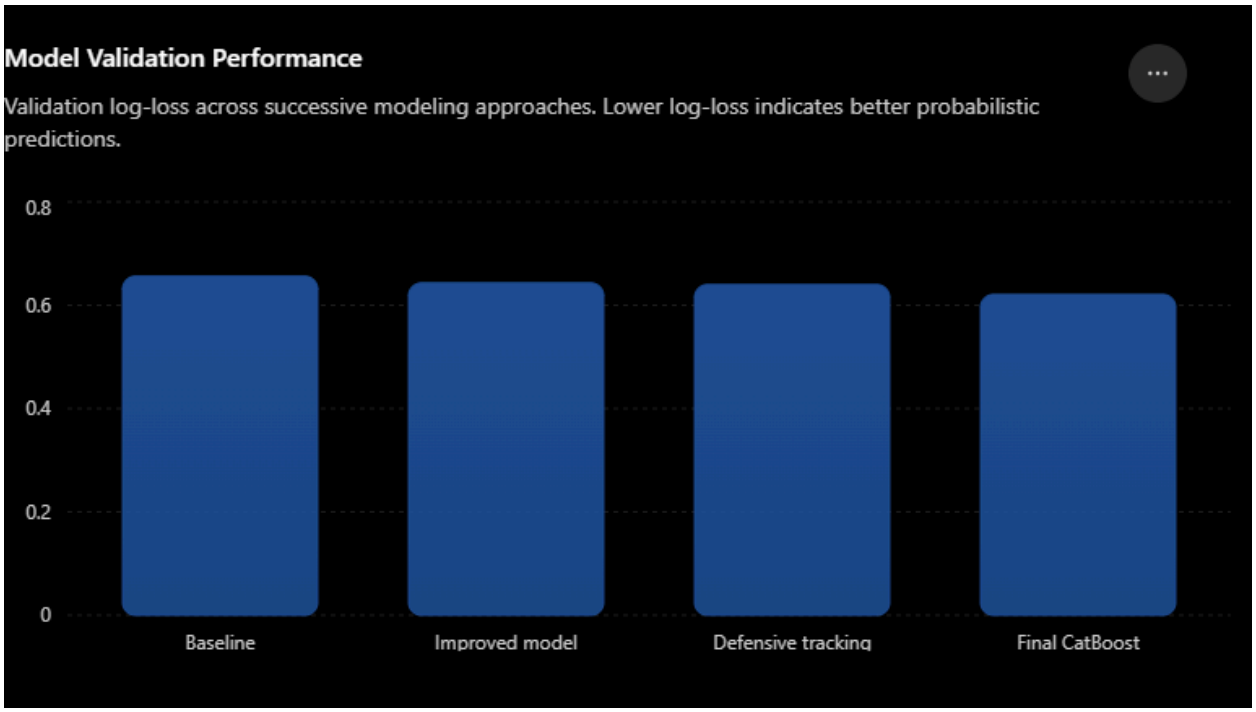
This was the best validation result obtained during the modeling process.

Model Performance

Model	Validation Log-Loss
Baseline	0.65618
Improved model	0.64317
Defensive-tracking model	0.63997

Final CatBoost	**0.62084**

The progression from the baseline to the final model represents a meaningful improvement in probabilistic prediction performance.



This visual is a validation log-loss across successive modeling approaches. Incorporating additional shot, contextual, and defensive-tracking features progressively improved probabilistic prediction performance.

Final Model and Feature Importance

The final CatBoost model incorporated the original shot, player, location, game-state, and defensive-tracking variables along with engineered features.

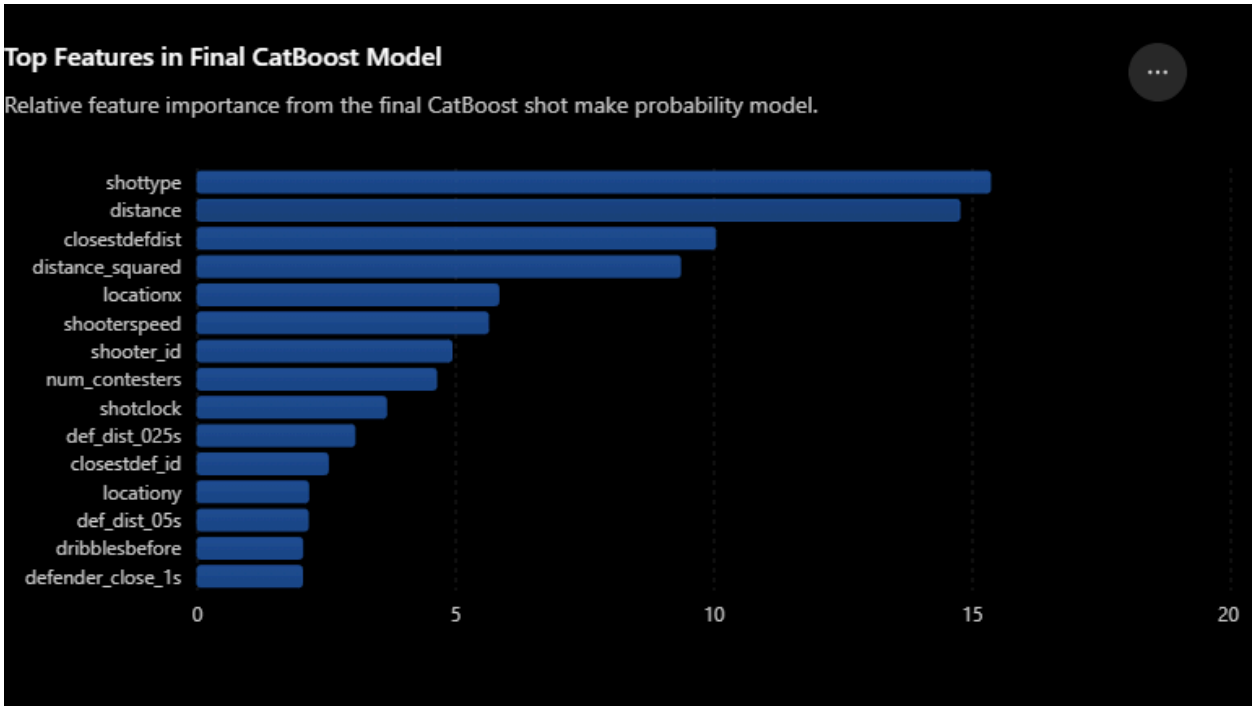
The most important features identified by the final model were:

Feature	Importance
Shot type	15.34
Distance	14.75
Closest defender distance	10.02

Distance squared	9.34
Location X	5.83
Shooter speed	5.63
Shooter ID	4.92
Number of contesters	4.62
Shot clock	3.65
Defender distance at 0.25 seconds	3.04
Closest defender ID	2.53
Location Y	2.14
Defender distance at 0.5 seconds	2.14
Dribbles before shot	2.03
Defender distance at 1 second	1.44

Shot type and shot distance were the two most influential features. Defensive proximity was also highly influential, with closest defender distance ranking third.

The importance of both shot characteristics and defensive-tracking features suggests that the model is incorporating information about both the inherent difficulty of the attempt and the defensive environment surrounding the shot.



This figure is the top 15 features by CatBoost feature importance. Shot type, shot distance, and closest defender distance were the most influential predictors in the final model.

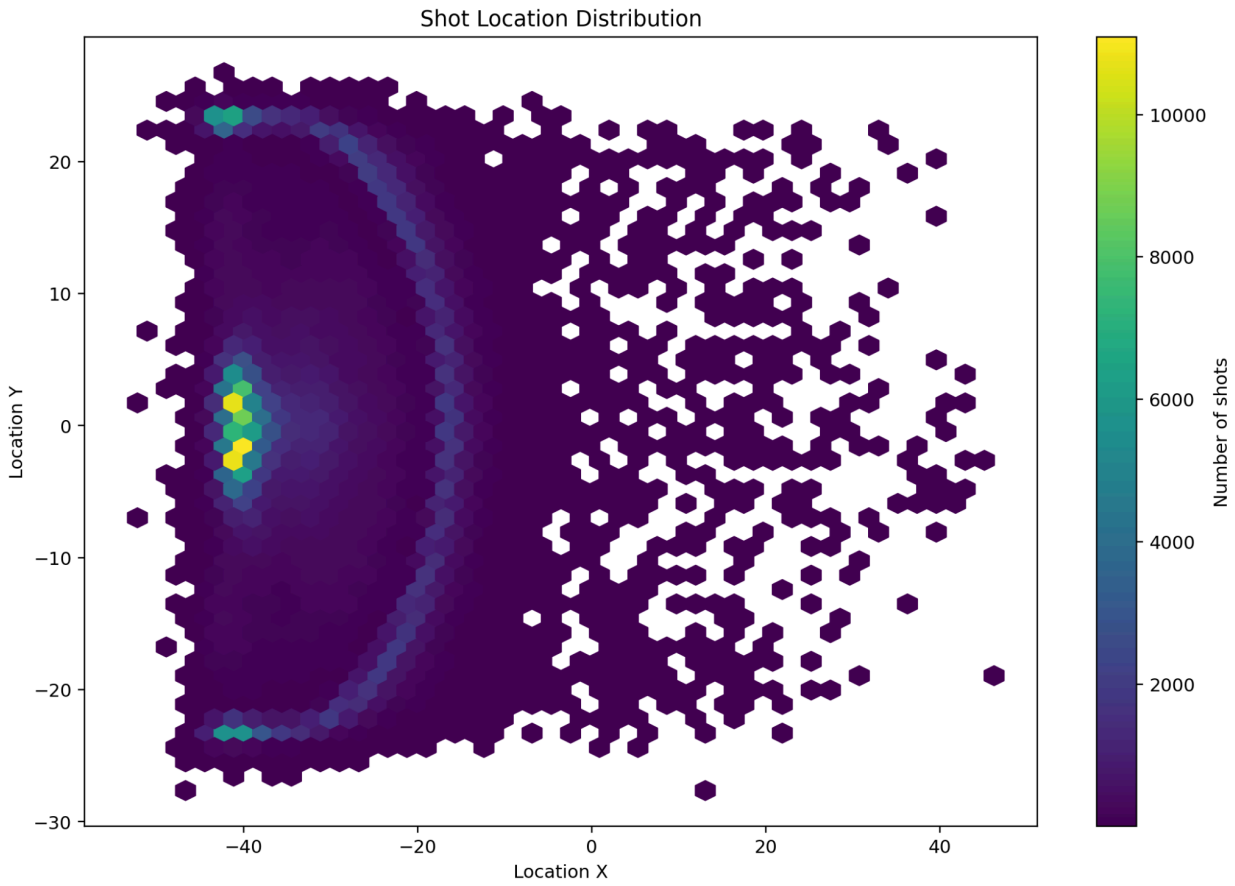
Defensive Tracking Analysis

A major part of the feature-engineering process was incorporating defensive tracking information.

The closest defender's distance is measured at several points leading up to the shot. Instead of treating the defensive information as a single value, I separated the measurements into features representing the defender's distance at different points before release.

This provides the model with information about the evolution of defensive pressure.

For example, two shots could have a similar defender distance at release while having very different defensive approaches immediately before release. Including multiple time points gives the model an opportunity to distinguish between these situations.



This visual details the distribution of shot locations in the training dataset. The spatial concentration of attempts provides additional information beyond shot distance alone.

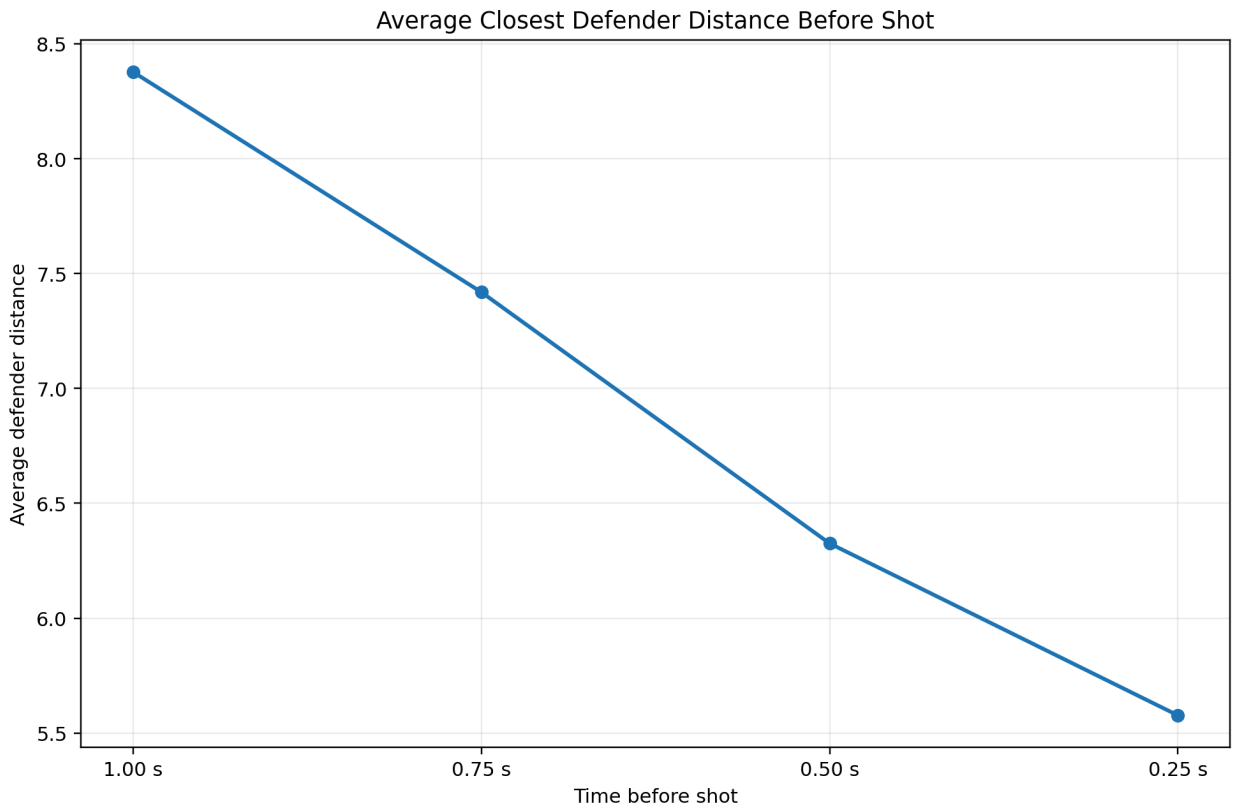
This feature group was also associated with an improvement in validation performance, supporting its inclusion in the final model.

Shot Location

Shot location was another important component of the model.

The dataset provides `locationx` and `locationy` coordinates describing where each shot occurred on the court. These variables allow the model to distinguish between attempts taken from different areas of the floor.

Shot distance captures the overall distance from the basket, while the location coordinates provide additional spatial information that can distinguish between different positions at similar distances.



This visual showcases the average closest-defender distance at four points preceding shot release. The decreasing distance illustrates how defensive pressure changes immediately before a shot and motivates the inclusion of defensive-tracking features in the model.

The combination of distance and spatial coordinates provides the model with a more complete representation of shot location.

Test-Set Predictions

After selecting the final model, I retrained it using the complete training dataset.

The final model was then used to generate make probabilities for all **213,977 shots** in the testing dataset.

The resulting `submission.csv` contains the required two columns:

shot_id,make_prob

Each row corresponds to one test shot and contains the model's estimated probability that the shot was made.

For example, a prediction such as:

1,0.3484

means the model estimated approximately a 34.8% probability that shot 1 would be made.

The predictions are probabilities rather than binary classifications, allowing the model to express different levels of confidence about shot difficulty.

Potential Basketball Applications

A shot make-probability model can provide useful information beyond simply identifying whether a shot was made.

For example, a shot predicted to have a 70% make probability is fundamentally different from one predicted to have a 25% probability, even though both ultimately result in either a make or miss.

A model of this type could potentially be used to:

- Evaluate shot difficulty
- Compare actual shooting results with expected results
- Identify players or situations that outperform or underperform expectations
- Analyze the effect of defensive pressure
- Evaluate how shot quality changes across game situations
- Support player and team-level analysis

The defensive-tracking features are particularly valuable because they provide information about shot difficulty that cannot be captured by location alone.

Limitations

There are several limitations to the current approach.

First, the model is limited to the variables available in the provided dataset. Additional information about offensive actions, defensive schemes, lineups, fatigue, player context, and other game circumstances could potentially improve predictions.

Second, feature importance describes predictive usefulness rather than causation. A feature being highly important to the model does not mean that changing that feature would directly cause the predicted probability to change by a particular amount.

Third, player and team identifiers may allow the model to learn historical differences between players and teams. While this can improve predictive performance, it can also make

generalization more difficult when the model encounters players or situations with limited historical data.

Finally, the validation result is based on the available validation split. Additional testing across different seasons or time periods would provide more information about how well the model generalizes to future situations.

Conclusion

This project developed a probabilistic NBA shot prediction model using shot characteristics, player information, spatial information, game context, and defensive-tracking data.

The modeling process began with a baseline log-loss of **0.65618** and progressively improved performance through additional feature engineering and model development.

The final CatBoost model achieved a validation log-loss of **0.62084** and generated make probabilities for all **213,977 test shots**.

The most influential features included shot type, shot distance, closest defender distance, spatial location, shooter speed, and defensive-tracking variables.

The results demonstrate how player-tracking and contextual information can be combined with machine learning to estimate shot outcomes probabilistically and provide a more detailed representation of shot difficulty than a simple make/miss classification.